Md Shamimul Islam

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Research Interests

Cybersecurity & Intrusion Detection | LLM & RAG | | Computer Vision for Safety Systems

Education

2025 – Present
- **MS in Computer Science, Florida Polytechnic University**
  CGPA: 3.77 out of 4.00

2020 – 2022
- **Masters in Applied Statistics and Data Science, Jahangirnagar University (JU)**
  CGPA: 3.96 out of 4.00
  Thesis title: *Video-based Bangla Sign Language Recognition using Deep learning Algorithms.*

2015 – 2020
- **B.Sc. Computer Science, Manarat International University (MIU).**
  CGPA: 3.99 out of 4.00
  Thesis title: *Pose-based Multi-level Features Fusion For Exam Cheat Recognition Using CNN.*

Research Publications

Peer-Reviewed Publications (Total Citation: 103 | h-index: 4 | i10-index: 2)

1. **M. S. Islam**, L. G. Jaimes, and A. S. Dina, "MA-IDS: Multi-agent RAG framework for IoT network intrusion detection with an experience library," in *Proceedings of the 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Accepted for publication, Bellevue, WA, USA, 2026. arXiv: 2604.05458 [cs.CR], (Citations: 5).

2. **M. S. Islam**, A. J. M. A. Joha, N. Hossain, M. M. Hasan, and S. Abdullah, "Multi-view bangla sign language recognition: A new word-level video dataset," in *2024 IEEE The 6th International Conference on Activity and Behavior Computing*, Nakatsu and Kitakyushu, Kyushu, Japan, 2024. ↗DOI: 10.1109/ABC61795.2024.10652018, (Citations: 7).

3. M. M. Hasan, **M. S. Islam**, and S. Abdullah, "Robust pose-based human fall detection using recurrent neural network," in *IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things*, Dhaka, Bangladesh, 2019. ↗DOI: 10.1109/RAAICON48939.2019.23, (Citations: 61).

4. **M. S. Islam**, M. M. Hasan, and S. Abdullah, "A deep spatio-temporal network for vision-based sexual harassment detection," in *IEEE International Conference on Emerging Technology in Computing, Communication and Electronics*, 2021. ↗DOI: 10.1109/ETCCE54784.2021.9689891, (Citations: 22).

5. **M. S. Islam** and I. Elwarfalli, "Deep learning-powered ecg-based biometric authentication," in *IEEE International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, DUET, Dhaka, Bangladesh, 2023. ↗DOI: 10.1109/NCIM59001.2023.10212845, (Citations: 4).

6. N. H. M. Arafat, S. Ileas, Mujazid, and **M. S. Islam**, "Blosecr: A conceptual framework of identity documents' security on blockchain in refugee issues and conserving fundamental rights," in *The International Conference for Intelligent Technologies*, Karnataka, India, 2022. ↗DOI: 10.1109/CONIT55038.2022.9847731, (Citations: 4).

Under Review / In Preparation

1. J. Al-Mahmud, **M. S. Islam**, and A. A. Jui, "Video-based sexual harassment detection using fine-tuned video transformers," Accepted to 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC), University of Dhaka, Bangladesh, 2026.
2. **M. S. Islam**, A. H. Sohag, and Z. Subah, "Permissionguard: Fine-tuned LLMs and BERT for android permission-based malware detection," Manuscript in preparation, 2025.

Datasets

1. A. J. M. A. Joha, M. N. Hossain, M. S. Islam, S. Abdullah, and M. M. Hasan, *Multi-View Bangla Sign Language (MV-BSL) Dataset*, GitHub repository, 2024. ↗URL: https://github.com/Mahedi-61/MV-BSDL.
2. S. Abdullah, M. S. Islam, and M. M. Hasan, *Bangladeshi Vehicle License Plate Dataset*, GitHub repository, https://github.com/Mahedi-61/Bangla_License_Plate_Dataset.
3. M. S. Islam and M. M. Hasan, *Sexual Harassment Video Dataset (SHVD)*, GitHub repository, https://github.com/Shamim-38/Sexual-Harassment-Video-Dataset-SHVD-.

Employment History

2025 – · · · ·	• **GA,** Department of Computer Science, Florida Polytechnic University.
2024 – 2025	• **Lecturer,** Department of Computer Science and Engineering (CSE), Manarat International University (MIU).
2023 – 2024	• **Lecturer,** Department of CSE, Asian University of Bangladesh (AUB) Convener, AUB Programming Club.
2022 – 2022	• **Adjunct Faculty,** Department of Islami Studies (DIS), MIU
2020 – 2022	• **Adjunct Faculty,** Department of CSE, MIU.

Research Projects

2025 – Present

- **MA-IDS: Multi-Agent RAG Framework for IoT Intrusion Detection (Thesis):** Developed a multi-agent IDS integrating GPT-4o with a FAISS-based Retrieval-Augmented Generation (RAG) library. Features a Traffic Classification Agent and an Error Analysis Agent for dynamic rule generation. Achieved Macro F1-Scores of 89.75% (NF-BoT-IoT) and 85.22% (NF-ToN-IoT), outperforming zero-shot baselines by up to 80%. Submitted to IEEE SMC 2026. Advisors: Dr. Ayesha S. Dina, Dr. Luis G. Jaimes (Florida Poly).
- **Video-Based Sexual Harassment Detection Using Fine-Tuned Video Transformers:** Extended the SHV dataset to 600 annotated clips across indoor and outdoor environments. Fine-tuned three state-of-the-art models (VideoMAE, ViViT, and TimeSformer) for binary classification. TimeSformer achieved the peak test accuracy of 92.27%, outperforming the traditional CNN-LSTM baseline of 88%. To be presented at IEEE 3Cs 2026.

Research Projects (continued)

2026 – Present
- **PermissionGuard: Fine-Tuned LLMs & BERT for Android Malware Detection:** Compared LoRA fine-tuning of LLaMA 3.1-8B and full fine-tuning of BERT-base against classical ML baselines for Android permission malware detection. LoRA achieved 97.00% (NATICUSdroid) and 88.39% (Mahindru) accuracy, exceeding baselines by 19+%. Course project; manuscript in preparation.

2024 – 2025
- **AI Career Counselor & Guidance Platform:** Built a LLaMA-powered platform providing personalized IT career roadmaps and labor market insights. Trained on an 8,000-point dataset curated from leading educational and job platforms.

2024
- **Detection of Phishing URLs for Secure Browsing:** Developed an ML system to detect fraudulent URLs using Logistic Regression, Random Forest, SVMs, and Neural Networks with advanced feature selection to optimize classification accuracy.

2022 – 2023
- **Real-time Bangla Sign Language Translation System:** Proposed a novel Transformer architecture to translate pose-based Bangla sign language video descriptors into sentences. Won the "Artificial Intelligence for Bangla Competition 2.0" (Short version).

2022
- **Bangla Social Network Post Analysis using Deep Learning:** Curated a custom dataset and trained a Transformer-based language model to classify political sentiment (pro/anti-government) and toxicity levels in real-time social media posts.

2018 – 2019
- **Automatic License Plate Recognition (ALPR):** Developed and deployed a real-time ALPR system trained on a Bangladeshi license plate dataset. Published and presented at an IEEE conference (2019).

Miscellaneous Experience

Awards and Achievements

2023
- **Winner of AI for Bangla 2.0 Competition:** Led a team to victory in the national-level competition directed by the Bangladesh Government, developing an innovative AI solution for Bangla sign language among 100+ competing teams.

2020
- **Islamic Development Bank (IsDB) Masters Scholarship:** Awarded the highly competitive scholarship ($6,000) for outstanding academic performance in graduate studies.

2018
- **University Merit Scholarship:** Awarded a full tuition waiver for placing in the top 1% of undergraduate students at Manarat International University.

Certification

2015
- **Champion,** Great Victory Day Essay Competition, Manarat International University.

Co/Extracurricular Activities

2026–Present
- **Vice President,** Muslim Students Association (MSA), Florida Polytechnic University, Lakeland, FL.
- **Research/Workshop Coordinator,** AI, Robotics, and Vision Research Club (AIRVRC), Florida Polytechnic University, Lakeland, FL.

Miscellaneous Experience (continued)

- 2026 • **Volunteer,** Summer Camp 2026, SALAM Academy Florida, FL. Assisted with community activities and student group management.
- 2019-2020 • **Elected President,** MIU CSE Club, Manarat International University, Ashulia, Bangladesh. Spearheaded department-wide initiatives and organized major events including MIU CSE Fest 2020 and the Intra-University Programming Contest.
- • **Organizing Member,** Code Camp, MIU CSE Club, Manarat International University, Ashulia, Bangladesh.
- 2019 • **Participant,** GDG Sonargaon Hackathon, Google Developer Group, Dhaka, Bangladesh.
- 2018 • **Participant,** ICPC Bangladesh National Collaborative Programming Contest (NCPC), IUBAT, Dhaka, Bangladesh.
- 2017 • **Co-founder,** Hasimukh (Smile), Rangpur, Bangladesh. Established a social welfare organization dedicated to supporting underprivileged communities.

Talks and Poster Presentations

- 2026 • **Invited Speaker,** 2026 Engineering and Science Student Research Symposium, Florida Polytechnic University, Lakeland, FL. Talk: "Adaptive Reasoning for Network Traffic Security: A Multi-Agent RAG Architecture with Persistent Knowledge Induction." Advisor: Dr. Ayesha S. Dina.
- • **Poster Presentation,** "MA-IDS: Multi-Agent RAG Framework for IoT Network Intrusion Detection with an Experience Library," Florida Polytechnic University, Lakeland, FL.
- 2022 • **Invited Speaker,** Webinar on "A Publication Road Map for Scientific Papers," organized by MIU CSE Club, MIU, Ashulia, Bangladesh.

Skills

Languages	• Strong reading, writing and speaking competencies for Bengali and English.
Coding	• Python, Java, PHP, C, R, SQL, XML/XSL.
System Administration	• Unix/Linux, Windows
Document Processing	• LaTeX, MS Office
Developer Tools	• Git, Google Cloud Platform
Libraries	• NumPy, OpenCV, Pandas, Matplotlib
AI Framework	• PyTorch, TensorFlow
Object Detection Algorithms	• YOLO, RetinaNet, Mask R-CNN
Pose Detection Algorithm	• OpenPose, MediaPipe
Misc.	• Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

References

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